USDAnalysis

March 5, 2026

[1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

[2]:
```
# Cell 2
file_path = "U.S._Chronic_Disease_Indicators.csv"
# First, load the data into a variable
raw = pd.read_csv(file_path, low_memory=False)  # Load data into 'raw' first

# Keep an untouched copy
usd_indicator = raw.copy()  # Now 'raw' is defined before using it

print("Loaded rows,cols:", usd_indicator.shape)
usd_indicator.head(5)
```

```
Loaded rows,cols: (309215, 30)
```

[2]:
```
   YearStart  YearEnd LocationAbbr   LocationDesc     DataSource  \
0       2020     2020           US  United States          BRFSS
1       2015     2019           AR       Arkansas  US Cancer DVT
2       2015     2019           CA     California  US Cancer DVT
3       2015     2019           CO       Colorado  US Cancer DVT
4       2015     2019           GA        Georgia  US Cancer DVT

           Topic                                          Question  Response  \
0  Health Status            Recent activity limitation among adults       NaN
1         Cancer    Invasive cancer (all sites combined), incidence       NaN
2         Cancer  Cervical cancer mortality among all females, u…       NaN
3         Cancer    Invasive cancer (all sites combined), incidence       NaN
4         Cancer  Prostate cancer mortality among all males, und…       NaN

  DataValueUnit      DataValueType  …  Unnamed: 20  Unnamed: 21  \
0        Number  Age-adjusted Mean  …          NaN          NaN
1        Number             Number  …          NaN          NaN
2        Number             Number  …          NaN          NaN
3        Number             Number  …          NaN          NaN
4        Number             Number  …          NaN          NaN
```

```
                                        Geolocation LocationID  TopicID  \
0                                               NaN         59      HEA
1    POINT (-92.27449074299966 34.74865012400045)          5      CAN
2   POINT (-120.99999953799971 37.63864012300047)          6      CAN
3  POINT (-106.13361092099967 38.843840757000464)          8      CAN
4    POINT (-83.62758034599966 32.83968109300048)         13      CAN

   QuestionID ResponseID DataValueTypeID  StratificationCategoryID1  \
0       HEA04        NaN      AGEADJMEAN                        SEX
1       CAN07        NaN            NMBR                        SEX
2       CAN03        NaN            NMBR                    OVERALL
3       CAN07        NaN            NMBR                       RACE
4       CAN05        NaN            NMBR                       RACE

   StratificationID1
0               SEXF
1               SEXM
2                OVR
3                HIS
4                WHT

[5 rows x 30 columns]
```

[3]:
```
usd_indicator.columns = usd_indicator.columns.str.strip()
```

[4]:
```
usd_indicator_updated = usd_indicator.loc[:, ~usd_indicator.columns.str.
 ↪contains("^Unnamed")]

print("Columns after cleanup:")
print(usd_indicator.columns)
```

```
Columns after cleanup:
Index(['YearStart', 'YearEnd', 'LocationAbbr', 'LocationDesc', 'DataSource',
       'Topic', 'Question', 'Response', 'DataValueUnit', 'DataValueType',
       'DataValue', 'DataValueAlt', 'DataValueFootnoteSymbol',
       'DataValueFootnote', 'LowConfidenceLimit', 'HighConfidenceLimit',
       'StratificationCategory1', 'Stratification1', 'Unnamed: 18',
       'Unnamed: 19', 'Unnamed: 20', 'Unnamed: 21', 'Geolocation',
       'LocationID', 'TopicID', 'QuestionID', 'ResponseID', 'DataValueTypeID',
       'StratificationCategoryID1', 'StratificationID1'],
      dtype='object')
```

[5]:
```
usd_indicator.head(3)
```

[5]:
```
   YearStart  YearEnd LocationAbbr   LocationDesc     DataSource  \
0       2020     2020           US  United States          BRFSS
```

```
1       2015     2019           AR       Arkansas  US Cancer DVT
2       2015     2019           CA     California  US Cancer DVT

           Topic                                          Question  Response  \
0  Health Status            Recent activity limitation among adults       NaN
1         Cancer    Invasive cancer (all sites combined), incidence       NaN
2         Cancer  Cervical cancer mortality among all females, u…       NaN

  DataValueUnit      DataValueType  …  Unnamed: 20  Unnamed: 21  \
0        Number  Age-adjusted Mean  …          NaN          NaN
1        Number             Number  …          NaN          NaN
2        Number             Number  …          NaN          NaN

                                     Geolocation LocationID  TopicID  \
0                                            NaN         59      HEA
1   POINT (-92.27449074299966 34.74865012400045)          5      CAN
2  POINT (-120.99999953799971 37.63864012300047)          6      CAN

   QuestionID ResponseID DataValueTypeID  StratificationCategoryID1  \
0       HEA04        NaN      AGEADJMEAN                        SEX
1       CAN07        NaN            NMBR                        SEX
2       CAN03        NaN            NMBR                    OVERALL

   StratificationID1
0               SEXF
1               SEXM
2                OVR

[3 rows x 30 columns]
```

[6]:
```
usd_indicator = usd_indicator.loc[:, ~usd_indicator.columns.str.
 ↪contains("Unnamed", case=False)]
```

[7]:
```
print(usd_indicator.columns.tolist())
```

```
['YearStart', 'YearEnd', 'LocationAbbr', 'LocationDesc', 'DataSource', 'Topic',
'Question', 'Response', 'DataValueUnit', 'DataValueType', 'DataValue',
'DataValueAlt', 'DataValueFootnoteSymbol', 'DataValueFootnote',
'LowConfidenceLimit', 'HighConfidenceLimit', 'StratificationCategory1',
'Stratification1', 'Geolocation', 'LocationID', 'TopicID', 'QuestionID',
'ResponseID', 'DataValueTypeID', 'StratificationCategoryID1',
'StratificationID1']
```

[8]:
```
usd_indicator.head()
```

[8]:
```
   YearStart  YearEnd LocationAbbr   LocationDesc     DataSource  \
0       2020     2020           US  United States          BRFSS
```

```
1       2015     2019           AR       Arkansas  US Cancer DVT
2       2015     2019           CA     California  US Cancer DVT
3       2015     2019           CO       Colorado  US Cancer DVT
4       2015     2019           GA        Georgia  US Cancer DVT

           Topic                                          Question  Response  \
0  Health Status            Recent activity limitation among adults       NaN
1         Cancer    Invasive cancer (all sites combined), incidence       NaN
2         Cancer  Cervical cancer mortality among all females, u…       NaN
3         Cancer    Invasive cancer (all sites combined), incidence       NaN
4         Cancer  Prostate cancer mortality among all males, und…       NaN

  DataValueUnit      DataValueType  …  StratificationCategory1  \
0        Number  Age-adjusted Mean  …                       Sex
1        Number             Number  …                       Sex
2        Number             Number  …                   Overall
3        Number             Number  …            Race/Ethnicity
4        Number             Number  …            Race/Ethnicity

      Stratification1                                     Geolocation  \
0              Female                                             NaN
1                Male    POINT (-92.27449074299966 34.74865012400045)
2             Overall   POINT (-120.99999953799971 37.63864012300047)
3            Hispanic  POINT (-106.13361092099967 38.843840757000464)
4  White, non-Hispanic    POINT (-83.62758034599966 32.83968109300048)

  LocationID  TopicID  QuestionID ResponseID DataValueTypeID  \
0         59      HEA       HEA04        NaN      AGEADJMEAN
1          5      CAN       CAN07        NaN            NMBR
2          6      CAN       CAN03        NaN            NMBR
3          8      CAN       CAN07        NaN            NMBR
4         13      CAN       CAN05        NaN            NMBR

  StratificationCategoryID1  StratificationID1
0                       SEX               SEXF
1                       SEX               SEXM
2                   OVERALL                OVR
3                      RACE                HIS
4                      RACE                WHT

[5 rows x 26 columns]
```

[9]:
```
resp_dropped = usd_indicator.drop(columns=["Response"])
```

[10]:
```
resp_dropped
```

[10]:

```
        YearStart  YearEnd LocationAbbr    LocationDesc     DataSource  \
0            2020     2020           US   United States          BRFSS
1            2015     2019           AR        Arkansas  US Cancer DVT
2            2015     2019           CA      California  US Cancer DVT
3            2015     2019           CO        Colorado  US Cancer DVT
4            2015     2019           GA         Georgia  US Cancer DVT
…             …        …            …              …             …
309210       2022     2022           VI  Virgin Islands          BRFSS
309211       2022     2022           WV   West Virginia          BRFSS
309212       2022     2022           WI       Wisconsin          BRFSS
309213       2022     2022           VT         Vermont          BRFSS
309214       2022     2022           WA      Washington          BRFSS

                                       Topic  \
0                               Health Status
1                                      Cancer
2                                      Cancer
3                                      Cancer
4                                      Cancer
…                                       …
309210                               Tobacco
309211  Chronic Obstructive Pulmonary Disease
309212                          Immunization
309213         Social Determinants of Health
309214                               Alcohol

                                                Question DataValueUnit  \
0                 Recent activity limitation among adults        Number
1         Invasive cancer (all sites combined), incidence        Number
2       Cervical cancer mortality among all females, u…        Number
3         Invasive cancer (all sites combined), incidence        Number
4       Prostate cancer mortality among all males, und…        Number
…                                                    …           …
309210  Quit attempts in the past year among adult cur…             %
309211  Chronic obstructive pulmonary disease among ad…             %
309212  Pneumococcal vaccination among adults aged 65 …             %
309213  Lack of health insurance among adults aged 18-…             %
309214             Binge drinking prevalence among adults             %

                  DataValueType  DataValue  …  StratificationCategory1  \
0             Age-adjusted Mean        2.9  …                       Sex
1                        Number     9537.0  …                       Sex
2                        Number      486.0  …                   Overall
3                        Number     2880.0  …            Race/Ethnicity
4                        Number      519.0  …            Race/Ethnicity
…                         …        …  …                         …
309210  Age-adjusted Prevalence        NaN  …            Race/Ethnicity
```

```
309211         Crude Prevalence       14.0  …                  Overall
309212         Crude Prevalence       64.2  …           Race/Ethnicity
309213         Crude Prevalence        NaN  …           Race/Ethnicity
309214  Age-adjusted Prevalence       19.2  …                      Sex

                                     Stratification1  \
0                                            Female
1                                              Male
2                                           Overall
3                                          Hispanic
4                                White, non-Hispanic
…                                                 …
309210  American Indian or Alaska Native, non-Hispanic
309211                                      Overall
309212                           Black, non-Hispanic
309213      Hawaiian or Pacific Islander, non-Hispanic
309214                                         Male

                                        Geolocation  LocationID  TopicID  \
0                                               NaN          59      HEA
1         POINT (-92.27449074299966 34.74865012400045)           5      CAN
2        POINT (-120.99999953799971 37.63864012300047)           6      CAN
3       POINT (-106.13361092099967 38.843840757000464)           8      CAN
4         POINT (-83.62758034599966 32.83968109300048)          13      CAN
…                                                   …         …       …
309210                   POINT (-64.896335 18.335765)          78      TOB
309211    POINT (-80.71264013499967 38.66551020200046)          54     COPD
309212    POINT (-89.81637074199966 44.39319117400049)          55      IMM
309213    POINT (-72.51764079099962 43.62538123900049)          50     SDOH
309214   POINT (-120.47001078999972 47.52227862900048)          53      ALC

       QuestionID ResponseID DataValueTypeID  StratificationCategoryID1  \
0           HEA04        NaN      AGEADJMEAN                        SEX
1           CAN07        NaN            NMBR                        SEX
2           CAN03        NaN            NMBR                    OVERALL
3           CAN07        NaN            NMBR                       RACE
4           CAN05        NaN            NMBR                       RACE
…           …        …              …                          …
309210      TOB06        NaN      AGEADJPREV                       RACE
309211     COPD01        NaN         CRDPREV                    OVERALL
309212      IMM04        NaN         CRDPREV                       RACE
309213      SDH09        NaN         CRDPREV                       RACE
309214      ALC06        NaN      AGEADJPREV                        SEX

       StratificationID1
0                   SEXF
1                   SEXM
```

```
2                    OVR
3                    HIS
4                    WHT
…                    …
309210              AIAN
309211               OVR
309212               BLK
309213              HAPI
309214              SEXM

[309215 rows x 25 columns]
```

[11]:
```
cleaned_data = usd_indicator.drop(columns=["ResponseID", "Response"],␣
↪errors="ignore")
```

[12]:
```
cleaned_data
```

[12]:
```
        YearStart  YearEnd LocationAbbr    LocationDesc     DataSource  \
0            2020     2020           US   United States          BRFSS
1            2015     2019           AR        Arkansas  US Cancer DVT
2            2015     2019           CA      California  US Cancer DVT
3            2015     2019           CO        Colorado  US Cancer DVT
4            2015     2019           GA         Georgia  US Cancer DVT
…             …        …            …               …             …
309210       2022     2022           VI  Virgin Islands          BRFSS
309211       2022     2022           WV   West Virginia          BRFSS
309212       2022     2022           WI       Wisconsin          BRFSS
309213       2022     2022           VT         Vermont          BRFSS
309214       2022     2022           WA      Washington          BRFSS

                                     Topic  \
0                             Health Status
1                                    Cancer
2                                    Cancer
3                                    Cancer
4                                    Cancer
…                                       …
309210                             Tobacco
309211  Chronic Obstructive Pulmonary Disease
309212                         Immunization
309213         Social Determinants of Health
309214                             Alcohol

                                          Question DataValueUnit  \
0                Recent activity limitation among adults        Number
1         Invasive cancer (all sites combined), incidence        Number
2       Cervical cancer mortality among all females, u…        Number
```

```
3         Invasive cancer (all sites combined), incidence        Number
4       Prostate cancer mortality among all males, und…        Number
…                                                   …           …
309210  Quit attempts in the past year among adult cur…             %
309211  Chronic obstructive pulmonary disease among ad…             %
309212  Pneumococcal vaccination among adults aged 65 …             %
309213  Lack of health insurance among adults aged 18-…             %
309214            Binge drinking prevalence among adults             %

                  DataValueType  DataValue  …  HighConfidenceLimit  \
0             Age-adjusted Mean        2.9  …                   2.9
1                        Number     9537.0  …                   NaN
2                        Number      486.0  …                   NaN
3                        Number     2880.0  …                   NaN
4                        Number      519.0  …                   NaN
…                          …         …  …                   …
309210  Age-adjusted Prevalence        NaN  …                   NaN
309211         Crude Prevalence       14.0  …                  15.2
309212         Crude Prevalence       64.2  …                  74.6
309213         Crude Prevalence        NaN  …                   NaN
309214  Age-adjusted Prevalence       19.2  …                  20.2

       StratificationCategory1  \
0                          Sex
1                          Sex
2                      Overall
3               Race/Ethnicity
4               Race/Ethnicity
…                          …
309210          Race/Ethnicity
309211                 Overall
309212          Race/Ethnicity
309213          Race/Ethnicity
309214                     Sex

                                      Stratification1  \
0                                             Female
1                                               Male
2                                            Overall
3                                           Hispanic
4                                White, non-Hispanic
…                                                 …
309210  American Indian or Alaska Native, non-Hispanic
309211                                       Overall
309212                           Black, non-Hispanic
309213     Hawaiian or Pacific Islander, non-Hispanic
309214                                          Male
```

```
                                          Geolocation  LocationID TopicID  \
0                                                NaN          59     HEA
1         POINT (-92.27449074299966 34.74865012400045)           5     CAN
2        POINT (-120.99999953799971 37.63864012300047)           6     CAN
3       POINT (-106.13361092099967 38.843840757000464)           8     CAN
4         POINT (-83.62758034599966 32.83968109300048)          13     CAN
…                                                  …         …      …
309210                    POINT (-64.896335 18.335765)          78     TOB
309211    POINT (-80.71264013499967 38.66551020200046)          54    COPD
309212    POINT (-89.81637074199966 44.39319117400049)          55     IMM
309213    POINT (-72.51764079099962 43.62538123900049)          50    SDOH
309214   POINT (-120.47001078999972 47.52227862900048)          53     ALC

       QuestionID DataValueTypeID  StratificationCategoryID1 StratificationID1
0           HEA04      AGEADJMEAN                        SEX              SEXF
1           CAN07            NMBR                        SEX              SEXM
2           CAN03            NMBR                    OVERALL               OVR
3           CAN07            NMBR                       RACE               HIS
4           CAN05            NMBR                       RACE               WHT
…           …              …                          …                …
309210      TOB06      AGEADJPREV                       RACE              AIAN
309211     COPD01         CRDPREV                    OVERALL               OVR
309212      IMM04         CRDPREV                       RACE               BLK
309213      SDH09         CRDPREV                       RACE              HAPI
309214      ALC06      AGEADJPREV                        SEX              SEXM

[309215 rows x 24 columns]
```

[13]:
```
# Drop confidence interval columns
df_cleaned_data = cleaned_data.drop(columns=["LowConfidenceLimit",␣
 ↪"HighConfidenceLimit"], errors="ignore")

# Confirm removal
print(df_cleaned_data.columns)
```

```
Index(['YearStart', 'YearEnd', 'LocationAbbr', 'LocationDesc', 'DataSource',
       'Topic', 'Question', 'DataValueUnit', 'DataValueType', 'DataValue',
       'DataValueAlt', 'DataValueFootnoteSymbol', 'DataValueFootnote',
       'StratificationCategory1', 'Stratification1', 'Geolocation',
       'LocationID', 'TopicID', 'QuestionID', 'DataValueTypeID',
       'StratificationCategoryID1', 'StratificationID1'],
      dtype='object')
```

[14]:
```
df_cleaned_data
```

[14]:

```
        YearStart  YearEnd LocationAbbr    LocationDesc     DataSource  \
0            2020     2020           US   United States          BRFSS
1            2015     2019           AR        Arkansas  US Cancer DVT
2            2015     2019           CA      California  US Cancer DVT
3            2015     2019           CO        Colorado  US Cancer DVT
4            2015     2019           GA         Georgia  US Cancer DVT
…             …        …            …              …             …
309210       2022     2022           VI  Virgin Islands          BRFSS
309211       2022     2022           WV   West Virginia          BRFSS
309212       2022     2022           WI       Wisconsin          BRFSS
309213       2022     2022           VT         Vermont          BRFSS
309214       2022     2022           WA      Washington          BRFSS

                                       Topic  \
0                               Health Status
1                                      Cancer
2                                      Cancer
3                                      Cancer
4                                      Cancer
…                                       …
309210                                Tobacco
309211  Chronic Obstructive Pulmonary Disease
309212                           Immunization
309213          Social Determinants of Health
309214                                Alcohol

                                                 Question DataValueUnit  \
0                 Recent activity limitation among adults        Number
1         Invasive cancer (all sites combined), incidence        Number
2       Cervical cancer mortality among all females, u…        Number
3         Invasive cancer (all sites combined), incidence        Number
4       Prostate cancer mortality among all males, und…        Number
…                                                    …           …
309210  Quit attempts in the past year among adult cur…             %
309211  Chronic obstructive pulmonary disease among ad…             %
309212  Pneumococcal vaccination among adults aged 65 …             %
309213  Lack of health insurance among adults aged 18-…             %
309214             Binge drinking prevalence among adults             %

                   DataValueType  DataValue  …  \
0             Age-adjusted Mean        2.9  …
1                        Number     9537.0  …
2                        Number      486.0  …
3                        Number     2880.0  …
4                        Number      519.0  …
…                         …        …  …
309210  Age-adjusted Prevalence        NaN  …
```

```
309211         Crude Prevalence       14.0  …
309212         Crude Prevalence       64.2  …
309213         Crude Prevalence        NaN  …
309214  Age-adjusted Prevalence       19.2  …

                                       DataValueFootnote  \
0                                                    NaN
1                                                    NaN
2                                                    NaN
3                                                    NaN
4                                                    NaN
…                                                    …
309210  No data available for this indicator because t…
309211                                               NaN
309212                                               NaN
309213  Data suppressed; denominator < 50 or relative …
309214                                               NaN

       StratificationCategory1  \
0                          Sex
1                          Sex
2                      Overall
3               Race/Ethnicity
4               Race/Ethnicity
…                          …
309210          Race/Ethnicity
309211                 Overall
309212          Race/Ethnicity
309213          Race/Ethnicity
309214                     Sex

                                     Stratification1  \
0                                             Female
1                                               Male
2                                            Overall
3                                           Hispanic
4                                White, non-Hispanic
…                                                …
309210  American Indian or Alaska Native, non-Hispanic
309211                                       Overall
309212                           Black, non-Hispanic
309213     Hawaiian or Pacific Islander, non-Hispanic
309214                                          Male

                                         Geolocation LocationID TopicID  \
0                                                NaN         59     HEA
1         POINT (-92.27449074299966 34.74865012400045)          5     CAN
```

```
2        POINT (-120.99999953799971 37.63864012300047)          6     CAN
3       POINT (-106.13361092099967 38.843840757000464)          8     CAN
4         POINT (-83.62758034599966 32.83968109300048)         13     CAN
…                                                   …        …     …
309210                   POINT (-64.896335 18.335765)         78     TOB
309211    POINT (-80.71264013499967 38.66551020200046)         54    COPD
309212    POINT (-89.81637074199966 44.39319117400049)         55     IMM
309213    POINT (-72.51764079099962 43.62538123900049)         50    SDOH
309214   POINT (-120.47001078999972 47.52227862900048)         53     ALC

        QuestionID DataValueTypeID StratificationCategoryID1 StratificationID1
0            HEA04      AGEADJMEAN                       SEX              SEXF
1            CAN07            NMBR                       SEX              SEXM
2            CAN03            NMBR                   OVERALL               OVR
3            CAN07            NMBR                      RACE               HIS
4            CAN05            NMBR                      RACE               WHT
…            …             …                         …                …
309210       TOB06      AGEADJPREV                      RACE              AIAN
309211      COPD01         CRDPREV                   OVERALL               OVR
309212       IMM04         CRDPREV                      RACE               BLK
309213       SDH09         CRDPREV                      RACE              HAPI
309214       ALC06      AGEADJPREV                       SEX              SEXM

[309215 rows x 22 columns]
```

[15]:
```
df_cleaned_data["DataValue"].isna().mean()
```

[15]:
```
0.3234610222660608
```

[16]:
```
df_cleaned_data_2 = cleaned_data.drop(columns=["DataValueFootnote",␣
↪"HighConfidenceLimit", "LowConfidenceLimit", "DataValue",␣
↪"DataValueFootnoteSymbol"], errors="ignore")
```

[17]:
```
df_cleaned_data_2
```

[17]:
```
        YearStart  YearEnd LocationAbbr    LocationDesc     DataSource  \
0            2020     2020           US   United States          BRFSS
1            2015     2019           AR        Arkansas  US Cancer DVT
2            2015     2019           CA      California  US Cancer DVT
3            2015     2019           CO        Colorado  US Cancer DVT
4            2015     2019           GA         Georgia  US Cancer DVT
…             …        …            …               …             …
309210       2022     2022           VI  Virgin Islands          BRFSS
309211       2022     2022           WV   West Virginia          BRFSS
309212       2022     2022           WI       Wisconsin          BRFSS
309213       2022     2022           VT         Vermont          BRFSS
309214       2022     2022           WA      Washington          BRFSS
```

```
                                        Topic  \
0                               Health Status
1                                      Cancer
2                                      Cancer
3                                      Cancer
4                                      Cancer
…                                        …
309210                                Tobacco
309211  Chronic Obstructive Pulmonary Disease
309212                           Immunization
309213          Social Determinants of Health
309214                                Alcohol

                                                Question DataValueUnit  \
0                 Recent activity limitation among adults        Number
1         Invasive cancer (all sites combined), incidence        Number
2       Cervical cancer mortality among all females, u…        Number
3         Invasive cancer (all sites combined), incidence        Number
4       Prostate cancer mortality among all males, und…        Number
…                                                     …            …
309210  Quit attempts in the past year among adult cur…              %
309211  Chronic obstructive pulmonary disease among ad…              %
309212  Pneumococcal vaccination among adults aged 65 …              %
309213  Lack of health insurance among adults aged 18-…              %
309214            Binge drinking prevalence among adults             %

                  DataValueType  DataValueAlt StratificationCategory1  \
0             Age-adjusted Mean           2.9                     Sex
1                        Number        9537.0                     Sex
2                        Number         486.0                 Overall
3                        Number        2880.0          Race/Ethnicity
4                        Number         519.0          Race/Ethnicity
…                         …           …                       …
309210  Age-adjusted Prevalence           NaN          Race/Ethnicity
309211         Crude Prevalence          14.0                 Overall
309212         Crude Prevalence          64.2          Race/Ethnicity
309213         Crude Prevalence           NaN          Race/Ethnicity
309214  Age-adjusted Prevalence          19.2                     Sex

                                       Stratification1  \
0                                            Female
1                                              Male
2                                           Overall
3                                          Hispanic
4                                White, non-Hispanic
…                                                 …
```

```
309210  American Indian or Alaska Native, non-Hispanic
309211                                         Overall
309212                             Black, non-Hispanic
309213      Hawaiian or Pacific Islander, non-Hispanic
309214                                            Male

                                        Geolocation  LocationID TopicID  \
0                                               NaN          59     HEA
1         POINT (-92.27449074299966 34.74865012400045)           5     CAN
2        POINT (-120.99999953799971 37.63864012300047)           6     CAN
3       POINT (-106.13361092099967 38.843840757000464)           8     CAN
4         POINT (-83.62758034599966 32.83968109300048)          13     CAN
…                                                   …         …     …
309210                   POINT (-64.896335 18.335765)          78     TOB
309211    POINT (-80.71264013499967 38.66551020200046)          54    COPD
309212    POINT (-89.81637074199966 44.39319117400049)          55     IMM
309213    POINT (-72.51764079099962 43.62538123900049)          50    SDOH
309214   POINT (-120.47001078999972 47.52227862900048)          53     ALC

       QuestionID DataValueTypeID StratificationCategoryID1 StratificationID1
0           HEA04      AGEADJMEAN                       SEX              SEXF
1           CAN07            NMBR                       SEX              SEXM
2           CAN03            NMBR                   OVERALL               OVR
3           CAN07            NMBR                      RACE               HIS
4           CAN05            NMBR                      RACE               WHT
…           …             …                         …                …
309210      TOB06      AGEADJPREV                      RACE              AIAN
309211     COPD01         CRDPREV                   OVERALL               OVR
309212      IMM04         CRDPREV                      RACE               BLK
309213      SDH09         CRDPREV                      RACE              HAPI
309214      ALC06      AGEADJPREV                       SEX              SEXM

[309215 rows x 19 columns]
```

[18]:
```
df_cleaned_data_2[df_cleaned_data_2["LocationAbbr"] == "US"].head()
```

[18]:
```
     YearStart  YearEnd LocationAbbr   LocationDesc     DataSource  \
0         2020     2020           US  United States          BRFSS
12        2015     2019           US  United States  US Cancer DVT
61        2018     2019           US  United States           NSCH
331       2019     2019           US  United States          YRBSS
332       2019     2019           US  United States          BRFSS

                     Topic  \
0             Health Status
12                   Cancer
61                    Sleep
```

```
331                  Alcohol
332  Cardiovascular Disease

                                                Question DataValueUnit  \
0                Recent activity limitation among adults        Number
12   Invasive cancer (all sites combined) mortality…        Number
61   Short sleep duration among children aged 4 mon…             %
331             Alcohol use among high school students             %
332  Taking medicine for high cholesterol among adults             %

         DataValueType  DataValueAlt StratificationCategory1 Stratification1  \
0    Age-adjusted Mean           2.9                     Sex          Female
12              Number      314982.0                     Sex            Male
61    Crude Prevalence          34.4                 Overall         Overall
331   Crude Prevalence          31.9                     Sex          Female
332   Crude Prevalence          28.0          Race/Ethnicity        Hispanic

    Geolocation  LocationID TopicID QuestionID DataValueTypeID  \
0           NaN          59     HEA      HEA04      AGEADJMEAN
12          NaN          59     CAN      CAN08            NMBR
61          NaN          59    SLEP      SLP01         CRDPREV
331         NaN          59     ALC      ALC01         CRDPREV
332         NaN          59     CVD      CVD04         CRDPREV

    StratificationCategoryID1 StratificationID1
0                         SEX              SEXF
12                        SEX              SEXM
61                    OVERALL               OVR
331                       SEX              SEXF
332                      RACE               HIS
```

[19]:
```
df_cleaned_data_2[df_cleaned_data_2["LocationAbbr"] == "US"].shape
```

[19]:
```
(5763, 19)
```

[20]:
```
usd_analysis = df_cleaned_data_2[df_cleaned_data_2["LocationAbbr"] != "US"].
 ↪copy()

print("Remaining rows:", usd_analysis.shape)
print("Unique locations:", usd_analysis["LocationAbbr"].nunique())
```

```
Remaining rows: (303452, 19)
Unique locations: 54
```

[21]:
```
usd_analysis
```

[21]:

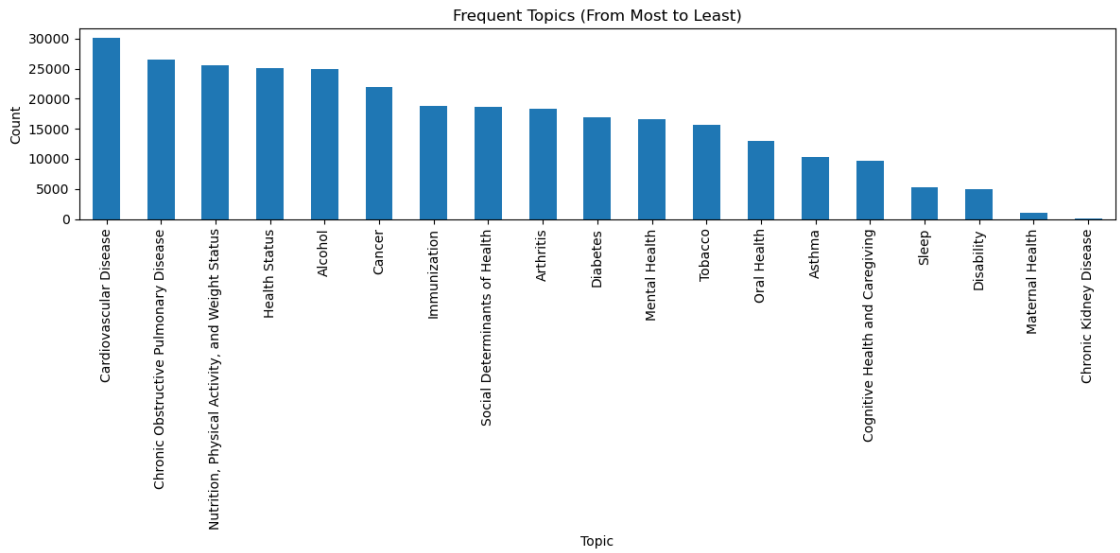
```
        YearStart  YearEnd LocationAbbr    LocationDesc     DataSource  \
1            2015     2019           AR        Arkansas  US Cancer DVT
2            2015     2019           CA      California  US Cancer DVT
3            2015     2019           CO        Colorado  US Cancer DVT
4            2015     2019           GA         Georgia  US Cancer DVT
5            2015     2019           KS          Kansas  US Cancer DVT
…             …        …            …             …             …
309210       2022     2022           VI  Virgin Islands          BRFSS
309211       2022     2022           WV   West Virginia          BRFSS
309212       2022     2022           WI       Wisconsin          BRFSS
309213       2022     2022           VT         Vermont          BRFSS
309214       2022     2022           WA      Washington          BRFSS

                                      Topic  \
1                                    Cancer
2                                    Cancer
3                                    Cancer
4                                    Cancer
5                                    Cancer
…                                     …
309210                             Tobacco
309211  Chronic Obstructive Pulmonary Disease
309212                         Immunization
309213         Social Determinants of Health
309214                             Alcohol

                                                Question DataValueUnit  \
1         Invasive cancer (all sites combined), incidence        Number
2       Cervical cancer mortality among all females, u…        Number
3         Invasive cancer (all sites combined), incidence        Number
4       Prostate cancer mortality among all males, und…        Number
5         Invasive cancer (all sites combined), incidence        Number
…                                                      …             …
309210  Quit attempts in the past year among adult cur…             %
309211  Chronic obstructive pulmonary disease among ad…             %
309212  Pneumococcal vaccination among adults aged 65 …             %
309213  Lack of health insurance among adults aged 18-…             %
309214            Binge drinking prevalence among adults             %

                  DataValueType  DataValueAlt StratificationCategory1  \
1                        Number        9537.0                     Sex
2                        Number         486.0                 Overall
3                        Number        2880.0          Race/Ethnicity
4                        Number         519.0          Race/Ethnicity
5                        Number        8102.0                     Sex
…                         …            …                       …
309210  Age-adjusted Prevalence           NaN          Race/Ethnicity
```

```
309211         Crude Prevalence          14.0                 Overall
309212         Crude Prevalence          64.2          Race/Ethnicity
309213         Crude Prevalence           NaN          Race/Ethnicity
309214  Age-adjusted Prevalence          19.2                     Sex

                                     Stratification1  \
1                                               Male
2                                            Overall
3                                           Hispanic
4                                White, non-Hispanic
5                                               Male
…                                                 …
309210  American Indian or Alaska Native, non-Hispanic
309211                                       Overall
309212                           Black, non-Hispanic
309213      Hawaiian or Pacific Islander, non-Hispanic
309214                                          Male

                                        Geolocation  LocationID TopicID  \
1         POINT (-92.27449074299966 34.74865012400045)           5     CAN
2        POINT (-120.99999953799971 37.63864012300047)           6     CAN
3       POINT (-106.13361092099967 38.843840757000464)           8     CAN
4         POINT (-83.62758034599966 32.83968109300048)          13     CAN
5         POINT (-98.20078122699965 38.34774030000045)          20     CAN
…                                                   …         …     …
309210                   POINT (-64.896335 18.335765)          78     TOB
309211    POINT (-80.71264013499967 38.66551020200046)          54    COPD
309212    POINT (-89.81637074199966 44.39319117400049)          55     IMM
309213    POINT (-72.51764079099962 43.62538123900049)          50    SDOH
309214   POINT (-120.47001078999972 47.52227862900048)          53     ALC

       QuestionID DataValueTypeID StratificationCategoryID1 StratificationID1
1           CAN07            NMBR                       SEX              SEXM
2           CAN03            NMBR                   OVERALL               OVR
3           CAN07            NMBR                      RACE               HIS
4           CAN05            NMBR                      RACE               WHT
5           CAN07            NMBR                       SEX              SEXM
…           …             …                         …                …
309210      TOB06      AGEADJPREV                      RACE              AIAN
309211     COPD01         CRDPREV                   OVERALL               OVR
309212      IMM04         CRDPREV                      RACE               BLK
309213      SDH09         CRDPREV                      RACE              HAPI
309214      ALC06      AGEADJPREV                       SEX              SEXM

[303452 rows x 19 columns]
```

[22]:
```
sorted(usd_analysis["Topic"].unique())
```

[22]:
```
['Alcohol',
 'Arthritis',
 'Asthma',
 'Cancer',
 'Cardiovascular Disease',
 'Chronic Kidney Disease',
 'Chronic Obstructive Pulmonary Disease',
 'Cognitive Health and Caregiving',
 'Diabetes',
 'Disability',
 'Health Status',
 'Immunization',
 'Maternal Health',
 'Mental Health',
 'Nutrition, Physical Activity, and Weight Status',
 'Oral Health',
 'Sleep',
 'Social Determinants of Health',
 'Tobacco']
```

[30]:
```
# Count records per topic
topic_counts = usd_analysis["Topic"].value_counts()

# Create single bar chart
plt.figure(figsize=(12,6))
topic_counts.plot(kind="bar")
plt.title("Frequent Topics (From Most to Least)")
plt.xlabel("Topic")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



[24]:
```
cvd_df = usd_analysis[usd_analysis["Topic"] == "Cardiovascular Disease"].copy()

print("Filtered shape:", cvd_df.shape)
```

```
Filtered shape: (30132, 19)
```

[25]:
```
cvd_questions = cvd_df["Question"].unique()

print("Number of questions under Cardiovascular Disease:", len(cvd_questions))
print(cvd_questions)
```

```
Number of questions under Cardiovascular Disease: 8
['Taking medicine to control high blood pressure among adults with high blood
pressure'
 'Coronary heart disease mortality among all people, underlying cause'
 'High cholesterol among adults who have been screened'
 'Taking medicine for high cholesterol among adults'
 'Cerebrovascular disease (stroke) mortality among all people, underlying cause'
 'High blood pressure among adults'
 'Diseases of the heart mortality among all people, underlying cause'
 'Hospitalization for heart failure as principal diagnosis, Medicare-
beneficiaries aged 65 years and older']
```

[26]:
```
copd_df = usd_analysis[usd_analysis["Topic"] == "Chronic Obstructive Pulmonary␣
↪Disease"].copy()

copd_questions = copd_df["Question"].unique()
```

```
print("Number of questions under Chronic Obstructive Pulmonary Disease:",␣
↪len(copd_questions))
print(copd_questions)
```

```
Number of questions under Chronic Obstructive Pulmonary Disease: 6
['Chronic obstructive pulmonary disease mortality among adults aged 45 years and
older, underlying cause'
 'Current smoking among adults with chronic obstructive pulmonary disease'
 'Chronic obstructive pulmonary disease among adults'
 'Chronic obstructive pulmonary disease mortality among adults aged 45 years and
older, underlying or contributing cause'
 'Hospitalization for chronic obstructive pulmonary disease as any diagnosis,
Medicare-beneficiaries aged 65 years and older'
 'Hospitalization for chronic obstructive pulmonary disease as principal
diagnosis, Medicare-beneficiaries aged 65 years and older']
```

[27]:
```
npw_df = usd_analysis[usd_analysis["Topic"] == "Nutrition, Physical Activity,␣
↪and Weight Status"].copy()

npw_questions = npw_df["Question"].unique()

print("Number of questions under Nutrition, Physical Activity, and Weight␣
↪Status:", len(npw_questions))
print(npw_questions)
```

```
Number of questions under Nutrition, Physical Activity, and Weight Status: 14
['Children and adolescents aged 6-13 years meeting aerobic physical activity
guideline'
 'Obesity among adults' 'No leisure-time physical activity among adults'
 'Consumed vegetables less than one time daily among adults'
 'Met aerobic physical activity guideline among high school students'
 'Consumed fruit less than one time daily among adults'
 'Consumed vegetables less than one time daily among high school students'
 'Obesity among high school students'
 'Consumed regular soda at least one time daily among high school students'
 'Met aerobic physical activity guideline for substantial health benefits,
adults'
 'Consumed fruit less than one time daily among high school students'
 'Infants who were breastfed at 12 months'
 'Infants who were exclusively breastfed through 6 months'
 'Obesity among WIC children aged 2 to 4 years']
```

[28]:
```
hstatus_df = usd_analysis[usd_analysis["Topic"] == "Health Status"].copy()

hstatus_questions = hstatus_df["Question"].unique()

print("Number of questions under Health Status:", len(hstatus_questions))
```

```
print(hstatus_questions)
```

```
Number of questions under Health Status: 6
['Frequent physical distress among adults'
 'Recent activity limitation among adults' 'Life expectancy at birth'
 'Fair or poor self-rated health status among adults'
 'Average recent physically unhealthy days among adults'
 '2 or more chronic conditions among adults']
```

[29]:
```
cancer_df = usd_analysis[usd_analysis["Topic"] == "Cancer"].copy()

cancer_questions = cancer_df["Question"].unique()

print("Number of questions under Cancer:", len(cancer_questions))
print(cancer_questions)
```

```
Number of questions under Cancer: 10
['Invasive cancer (all sites combined), incidence'
 'Cervical cancer mortality among all females, underlying cause'
 'Prostate cancer mortality among all males, underlying cause'
 'Breast cancer mortality among all females, underlying cause'
 'Lung and bronchial cancer mortality among all people, underlying cause'
 'Invasive cancer (all sites combined) mortality among all people, underlying
cause'
 'Colon and rectum (colorectal) cancer mortality among all people, underlying
cause'
 'Mammography use among women aged 50-74 years'
 'Cervical cancer screening among women aged 21-65 years'
 'Colorectal cancer screening among adults aged 45-75 years']
```

[39]:
```
diabetes_df = usd_analysis[usd_analysis["Topic"] == "Diabetes"].copy()

diabetes_questions = diabetes_df["Question"].unique()

print("Number of questions under Diabetes:", len(diabetes_questions))
print(diabetes_questions)
```

```
Number of questions under Diabetes: 4
['Diabetic ketoacidosis mortality among all people, underlying or contributing
cause'
 'Diabetes among adults'
 'Gestational diabetes among women with a recent live birth'
 'Diabetes mortality among all people, underlying or contributing cause']
```

0.0.1 Let's now group U.S. states into distinct health profiles using chronic disease indicators ****

Here we will use Clustering Strategy:

[43]:
```
usd_analysis.dtypes
```

[43]:
```
YearStart                      int64
YearEnd                        int64
LocationAbbr                  object
LocationDesc                  object
DataSource                    object
Topic                         object
Question                      object
DataValueUnit                 object
DataValueType                 object
DataValueAlt                 float64
StratificationCategory1       object
Stratification1               object
Geolocation                   object
LocationID                     int64
TopicID                       object
QuestionID                    object
DataValueTypeID               object
StratificationCategoryID1     object
StratificationID1             object
dtype: object
```

[46]:
```
print(df_cleaned_data.columns)
```

```
Index(['YearStart', 'YearEnd', 'LocationAbbr', 'LocationDesc', 'DataSource',
       'Topic', 'Question', 'DataValueUnit', 'DataValueType', 'DataValue',
       'DataValueAlt', 'DataValueFootnoteSymbol', 'DataValueFootnote',
       'StratificationCategory1', 'Stratification1', 'Geolocation',
       'LocationID', 'TopicID', 'QuestionID', 'DataValueTypeID',
       'StratificationCategoryID1', 'StratificationID1'],
      dtype='object')
```

0.0.2 Below code filters our variable 'usd_analysis' and adds a missing column 'DataValue'

[51]:
```
# Cell 1
# Use df_cleaned as the canonical cleaned long-form dataframe
# (replace df_cleaned with the correct variable name in your notebook if␣
↪different)
base = globals().get("df_cleaned_data")
if base is None:
    raise RuntimeError("df_cleaned not found - make sure you have the cleaned␣
↪long dataframe loaded.")

# Make a copy to avoid accidental in-place edits
usd_analysis = base.copy()
```

```
# Ensure column names are clean
usd_analysis.columns = usd_analysis.columns.str.strip()

# Remove national aggregate rows (US) if present
if "LocationAbbr" in usd_analysis.columns:
    usd_analysis = usd_analysis[usd_analysis["LocationAbbr"] != "US"].copy()

# Keep percent-type indicators only (adjust if you want different unit)
if "DataValueUnit" in usd_analysis.columns:
    usd_analysis = usd_analysis[usd_analysis["DataValueUnit"].str.contains("%",␣
 ↪na=False)].copy()

# Keep overall stratification only if available
if "Stratification1" in usd_analysis.columns and "Overall" in␣
 ↪usd_analysis["Stratification1"].unique():
    usd_analysis = usd_analysis[usd_analysis["Stratification1"] == "Overall"].
 ↪copy()
```

[52]:
```
print(usd_analysis.columns)
```

```
Index(['YearStart', 'YearEnd', 'LocationAbbr', 'LocationDesc', 'DataSource',
       'Topic', 'Question', 'DataValueUnit', 'DataValueType', 'DataValue',
       'DataValueAlt', 'DataValueFootnoteSymbol', 'DataValueFootnote',
       'StratificationCategory1', 'Stratification1', 'Geolocation',
       'LocationID', 'TopicID', 'QuestionID', 'DataValueTypeID',
       'StratificationCategoryID1', 'StratificationID1'],
      dtype='object')
```

[56]:
```
# Treat DataValue as values
wide = usd_analysis.pivot_table(
    index=["LocationDesc", "YearStart", "YearEnd"],
    columns="Question",
    values="DataValue"
).reset_index()

# clean column names (safe identifiers)
wide.columns = [str(c).strip().replace(" ", "_").replace("/", "_").replace("-",␣
 ↪"_") for c in wide.columns]
```

[57]:
```
# Cell 3
feature_cols = [c for c in wide.columns if c not in ("LocationDesc",␣
 ↪"YearStart", "YearEnd")]
for c in feature_cols:
    wide[c] = pd.to_numeric(wide[c], errors="coerce")

# check for any non-numeric or all-NaN columns
```

```
non_numeric_after = [c for c in feature_cols if not pd.api.types.
 ↪is_numeric_dtype(wide[c])]
all_nan = [c for c in feature_cols if wide[c].isna().all()]
print("Non-numeric after coercion (should be empty):", non_numeric_after)
print("Columns all NaN (consider dropping):", len(all_nan))
if len(all_nan)>0:
    print("Sample all-NaN columns:", all_nan[:10])
```

```
Non-numeric after coercion (should be empty): []
Columns all NaN (consider dropping): 0
```

[59]:
```
# Cell 5
missing_pct = state_df.isna().mean().sort_values(ascending=False)
print("Top missing cols:\n", missing_pct.head(10))

# Drop columns with > 40% missing (tune threshold as desired)
threshold = 0.40
keep_cols = missing_pct[missing_pct <= threshold].index.tolist()
if "LocationDesc" not in keep_cols:
    keep_cols = ["LocationDesc"] + keep_cols
state_df = state_df[keep_cols].copy()
print("After dropping sparse cols, shape:", state_df.shape)
```

```
Top missing cols:
 Current_poor_mental_health_among_high_school_students
0.351852
Unable_to_pay_mortgage,_rent,_or_utility_bills_in_the_past_12_months_among_adult
s                 0.314815
Lack_of_reliable_transportation_in_the_past_12_months_among_adults
0.314815
Lack_of_social_and_emotional_support_needed_among_adults
0.314815
Consumed_vegetables_less_than_one_time_daily_among_high_school_students
0.296296
Current_tobacco_use_of_any_tobacco_product_among_high_school_students
0.240741
Current_smokeless_tobacco_use_among_high_school_students
0.203704
Consumed_regular_soda_at_least_one_time_daily_among_high_school_students
0.185185
Preventive_dental_care_in_the_12_months_before_pregnancy_among_women_with_a_rece
nt_live_birth    0.166667
Health_insurance_coverage_in_the_month_before_pregnancy_among_women_with_a_recen
t_live_birth     0.166667
dtype: float64
After dropping sparse cols, shape: (54, 85)
```

[60]:
```
# Cell 6
# Ensure only one non-numeric column (LocationDesc)
print(state_df.dtypes.value_counts())
print("Any remaining non-numeric columns:", [c for c in state_df.columns if not␣
↪pd.api.types.is_numeric_dtype(state_df[c])])

# Final feature matrix for clustering
X = state_df.drop(columns=["LocationDesc"])
print("X shape:", X.shape)
```

```
float64    84
object      1
Name: count, dtype: int64
Any remaining non-numeric columns: ['LocationDesc']
X shape: (54, 84)
```

[67]:
```
# drop temporal columns before clustering
state_df = state_df.drop(columns=["YearStart","YearEnd"], errors="ignore")
```

[70]:
```
# Option A: select questions by Topic via the long df (recommended)
topics_of_interest = [
    "Cardiovascular Disease",
    "Chronic Kidney Disease",
    "Chronic Obstructive Pulmonary Disease",
    "Nutrition, Physical Activity, and Weight Status",
    "Health Status",
    "Cancer",
    "Diabetes"
]

# get unique (Topic, Question) pairs from the long cleaned frame
topic_question = (
    df_cleaned_data[["Topic","Question"]]
    .drop_duplicates()
)

# list all Questions for chosen topics
questions_sel = topic_question[topic_question["Topic"].
↪isin(topics_of_interest)]["Question"].unique().tolist()
print("Questions found for selected topics:", len(questions_sel))

# show a sample
print(questions_sel[:40])

# Now pick the corresponding wide/state_df columns (they may have different␣
↪chars)
```

```python
# build a mapping from original question text -> safe column name used in␣
↪state_df (if you cleaned columns earlier)
def safe_name(s):
    return str(s).strip().replace(" ", "_").replace("/", "_").replace("-", "_")

safe_map = {safe_name(q): q for q in questions_sel}
# find which of those safe names are present in state_df
present = [c for c in state_df.columns if c in safe_map]
missing = [c for c in safe_map.keys() if c not in state_df.columns]

print("Present question-columns in state_df:", len(present))
print("Missing (not found) sample:", missing[:10])

# select the ones present
features = present.copy()
cluster_df = state_df[["LocationDesc"] + features].copy()
print("Selected features:", len(features))
```

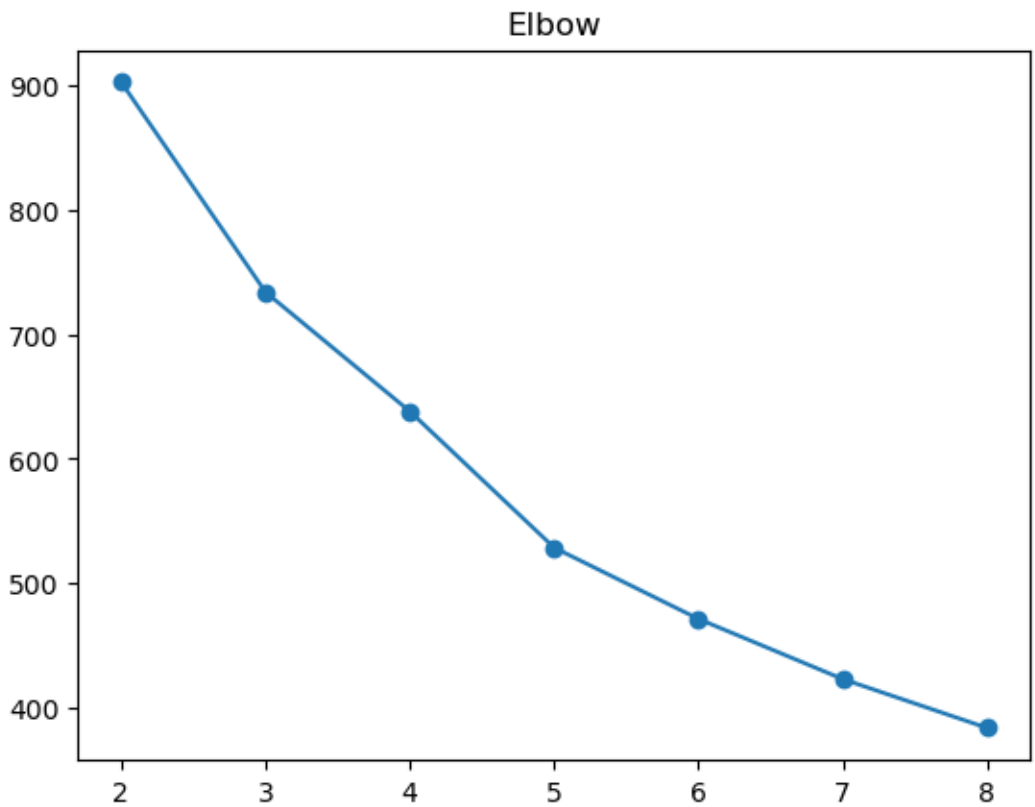
```
Questions found for selected topics: 49
['Recent activity limitation among adults', 'Invasive cancer (all sites
combined), incidence', 'Cervical cancer mortality among all females, underlying
cause', 'Prostate cancer mortality among all males, underlying cause', 'Breast
cancer mortality among all females, underlying cause', 'Invasive cancer (all
sites combined) mortality among all people, underlying cause', 'Diabetic
ketoacidosis mortality among all people, underlying or contributing cause',
'Lung and bronchial cancer mortality among all people, underlying cause', 'Colon
and rectum (colorectal) cancer mortality among all people, underlying cause',
'Children and adolescents aged 6-13 years meeting aerobic physical activity
guideline', 'Taking medicine to control high blood pressure among adults with
high blood pressure', 'Coronary heart disease mortality among all people,
underlying cause', 'Chronic obstructive pulmonary disease mortality among adults
aged 45 years and older, underlying cause', 'Frequent physical distress among
adults', 'Obesity among adults', 'Current smoking among adults with chronic
obstructive pulmonary disease', 'High cholesterol among adults who have been
screened', 'Taking medicine for high cholesterol among adults', 'Cerebrovascular
disease (stroke) mortality among all people, underlying cause', 'High blood
pressure among adults', 'Diseases of the heart mortality among all people,
underlying cause', 'Diabetes among adults', 'Gestational diabetes among women
with a recent live birth', 'Diabetes mortality among all people, underlying or
contributing cause', 'Life expectancy at birth', 'Fair or poor self-rated health
status among adults', 'Chronic obstructive pulmonary disease among adults', 'No
leisure-time physical activity among adults', 'Consumed vegetables less than one
time daily among adults', 'Met aerobic physical activity guideline among high
school students', 'Incidence of treated end-stage kidney disease', 'Consumed
fruit less than one time daily among adults', 'Average recent physically
unhealthy days among adults', 'Consumed vegetables less than one time daily
among high school students', 'Chronic obstructive pulmonary disease mortality
```

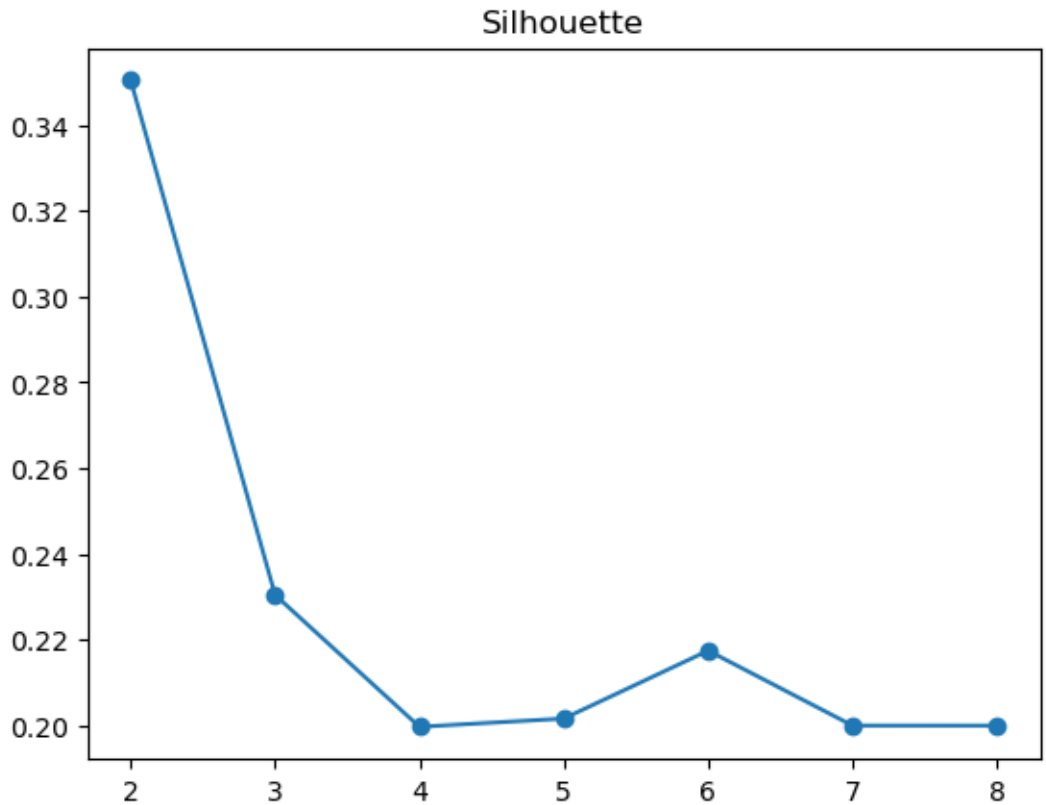
```
among adults aged 45 years and older, underlying or contributing cause', '2 or
more chronic conditions among adults', 'Mammography use among women aged 50-74
years', 'Cervical cancer screening among women aged 21-65 years', 'Colorectal
cancer screening among adults aged 45-75 years', 'Hospitalization for chronic
obstructive pulmonary disease as any diagnosis, Medicare-beneficiaries aged 65
years and older']
Present question-columns in state_df: 28
Missing (not found) sample: ['Recent_activity_limitation_among_adults',
'Invasive_cancer_(all_sites_combined),_incidence',
'Cervical_cancer_mortality_among_all_females,_underlying_cause',
'Prostate_cancer_mortality_among_all_males,_underlying_cause',
'Breast_cancer_mortality_among_all_females,_underlying_cause', 'Invasive_cancer_
(all_sites_combined)_mortality_among_all_people,_underlying_cause', 'Diabetic_ke
toacidosis_mortality_among_all_people,_underlying_or_contributing_cause',
'Lung_and_bronchial_cancer_mortality_among_all_people,_underlying_cause', 'Colon
_and_rectum_(colorectal)_cancer_mortality_among_all_people,_underlying_cause',
'Coronary_heart_disease_mortality_among_all_people,_underlying_cause']
Selected features: 28
```

[71]:
```python
# standard pre-processing and clustering
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt

# impute with median
num_feats = [c for c in cluster_df.columns if c!="LocationDesc"]
imp = SimpleImputer(strategy="median")
X = imp.fit_transform(cluster_df[num_feats])

# scale
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# PCA to reduce noise (optional)
pca = PCA(n_components=0.90, svd_solver="full")  # keep 90% variance
X_pca = pca.fit_transform(X_scaled)
print("PCA components:", X_pca.shape[1])

# elbow & silhouette
inertia, sil = [], []
K = range(2,9)
for k in K:
    km = KMeans(n_clusters=k, random_state=42, n_init=20).fit(X_pca)
    inertia.append(km.inertia_)
```

```
    sil.append(silhouette_score(X_pca, km.labels_))

plt.figure(); plt.plot(K, inertia, '-o'); plt.title("Elbow"); plt.show()
plt.figure(); plt.plot(K, sil, '-o'); plt.title("Silhouette"); plt.show()
print("silhouette scores:", dict(zip(K, sil)))
```

```
PCA components: 9
```






```
silhouette scores: {2: 0.35043844689886566, 3: 0.23043088783835405, 4:
0.19967255275820148, 5: 0.20159245966902062, 6: 0.21743644089906797, 7:
0.1999639734768836, 8: 0.19995333739364926}
```

[73]:
```python
# Now compute cluster summary from the original features (safe and␣
↪interpretable)
num_feats = [c for c in cluster_df.columns if c not in ("LocationDesc",␣
↪"Cluster")]
cluster_summary = cluster_df.groupby("Cluster")[num_feats].mean()
print(cluster_summary.T)   # transpose for easier reading (features down,␣
↪clusters across)
```

```
Cluster                                                    0          1
Consumed_vegetables_less_than_one_time_daily_am…  12.913636   8.194444
Consumed_regular_soda_at_least_one_time_daily_a…  22.945455  14.151515
Gestational_diabetes_among_women_with_a_recent_…   9.665000  10.651905
Consumed_fruit_less_than_one_time_daily_among_h…  10.995833   7.616667
Obesity_among_high_school_students                  19.041667  15.295946
Met_aerobic_physical_activity_guideline_among_h…  22.800000  22.891892
Children_and_adolescents_aged_6_13_years_meetin…  26.995455  26.413750
Infants_who_were_exclusively_breastfed_through_…  22.183333  27.458750
```

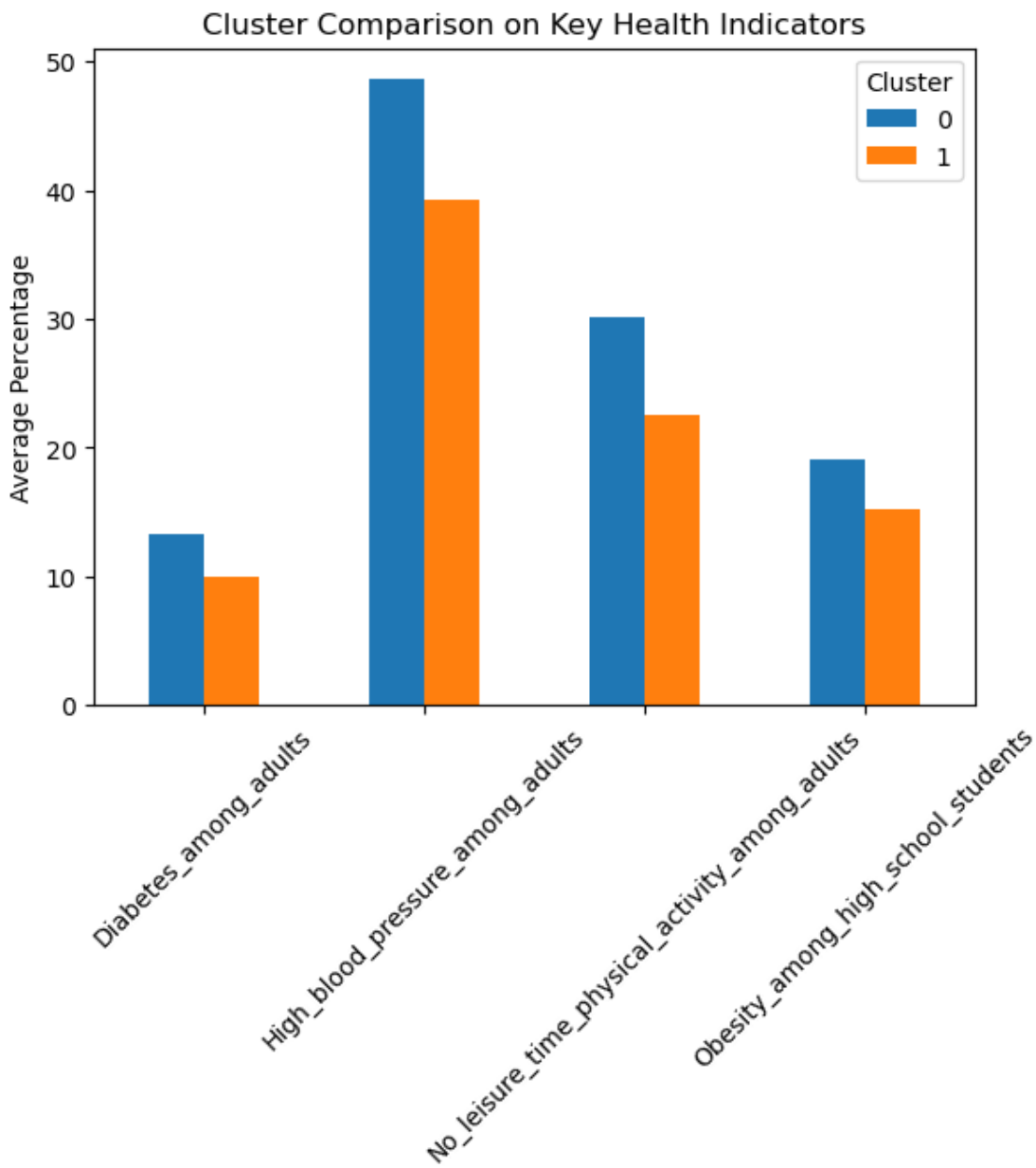
```
Infants_who_were_breastfed_at_12_months              28.987500  39.677500
Met_aerobic_physical_activity_guideline_for_sub…  43.625000  53.090000
Cervical_cancer_screening_among_women_aged_21_6…  83.312500  82.847561
Current_smoking_among_adults_with_chronic_obstr…  40.465278  36.063415
Taking_medicine_to_control_high_blood_pressure_…  73.308333  66.020833
Taking_medicine_for_high_cholesterol_among_adults   34.589583  30.916667
2_or_more_chronic_conditions_among_adults            20.943750  15.893750
High_cholesterol_among_adults_who_have_been_scr…  34.906250  31.180357
Frequent_physical_distress_among_adults              13.832292  10.656052
High_blood_pressure_among_adults                     48.579167  39.270238
Diabetes_among_adults                                13.310417   9.989980
Mammography_use_among_women_aged_50_74_years         74.568750  75.250000
No_leisure_time_physical_activity_among_adults       30.117708  22.573611
Obesity_among_WIC_children_aged_2_to_4_years         13.983333  13.861905
Fair_or_poor_self_rated_health_status_among_adults  20.830208  15.033433
Consumed_vegetables_less_than_one_time_daily_am…  22.870833  20.119643
Obesity_among_adults                                 37.401042  31.649802
Chronic_obstructive_pulmonary_disease_among_adults   8.712500   5.641964
Colorectal_cancer_screening_among_adults_aged_4…  60.637500  60.491071
Consumed_fruit_less_than_one_time_daily_among_a…  45.910417  39.242262
```

[74]:
```python
cluster_df[cluster_df["Cluster"] == 0]["LocationDesc"]
```

[74]:
```
0            Alabama
3           Arkansas
10           Georgia
15           Indiana
18          Kentucky
19         Louisiana
25       Mississippi
37          Oklahoma
40       Puerto Rico
42    South Carolina
44         Tennessee
51     West Virginia
Name: LocationDesc, dtype: object
```

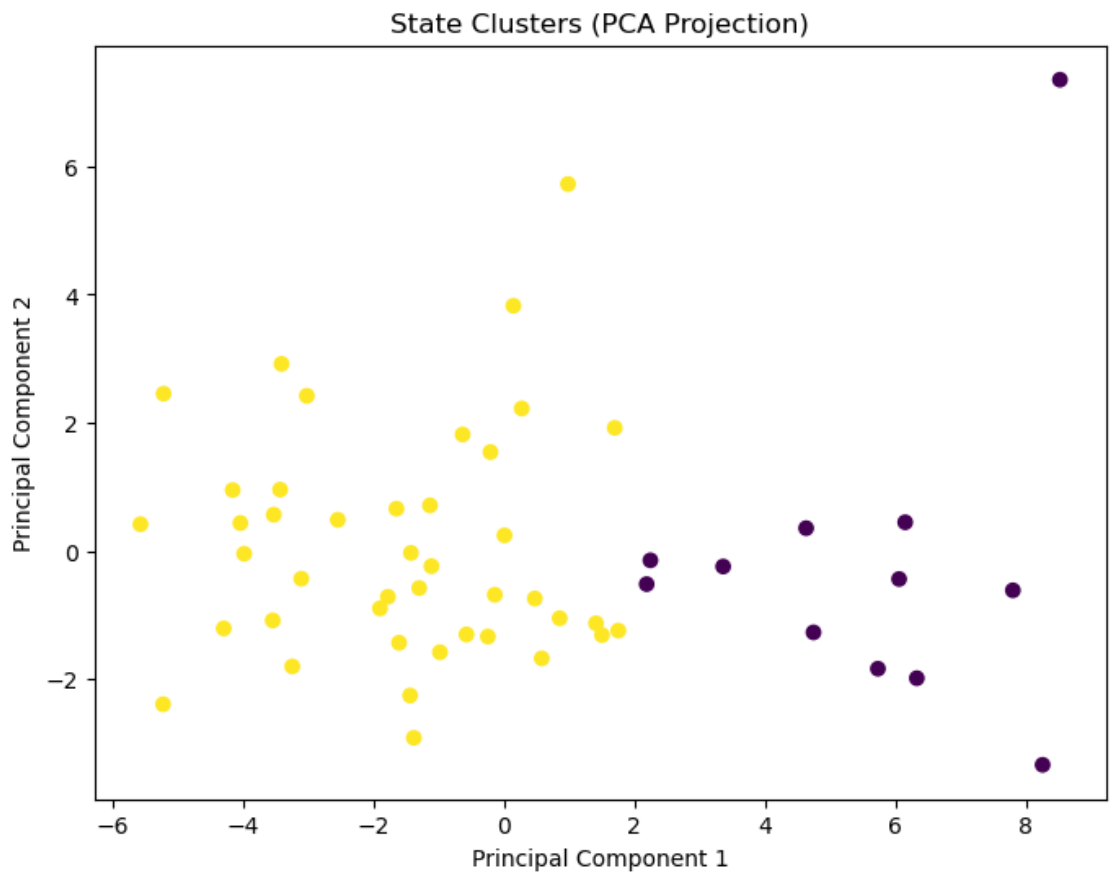
[79]:
```python
cluster_summary = cluster_df.groupby("Cluster")[num_feats].mean()

# Select a few key indicators
important = [
    "Diabetes_among_adults",
    "High_blood_pressure_among_adults",
    "No_leisure_time_physical_activity_among_adults",
    "Obesity_among_high_school_students"
]
```

```
cluster_summary[important].T.plot(kind="bar")
plt.title("Cluster Comparison on Key Health Indicators")
plt.ylabel("Average Percentage")
plt.xticks(rotation=45)
plt.show()
```



[80]:
```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
```

```
pca = PCA(n_components=2)
X_2d = pca.fit_transform(X_scaled)

plt.figure(figsize=(8,6))
plt.scatter(X_2d[:,0], X_2d[:,1], c=cluster_df["Cluster"])
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("State Clusters (PCA Projection)")
plt.show()
```



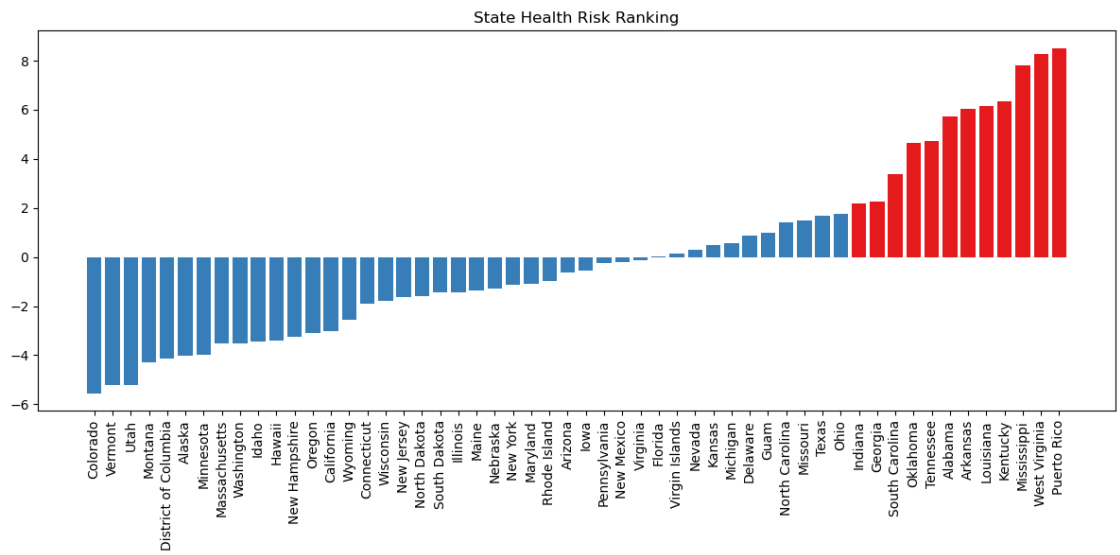
[81]:
```
high_risk_cluster = 0  # change if needed

high_risk_states = cluster_df[cluster_df["Cluster"] ==␣
↪high_risk_cluster]["LocationDesc"]

print("High-Risk States:")
print(high_risk_states.tolist())
```

```
High-Risk States:
['Alabama', 'Arkansas', 'Georgia', 'Indiana', 'Kentucky', 'Louisiana',
```

'Mississippi', 'Oklahoma', 'Puerto Rico', 'South Carolina', 'Tennessee', 'West Virginia']

0.0.3 High-Risk Vs Low-Risk States

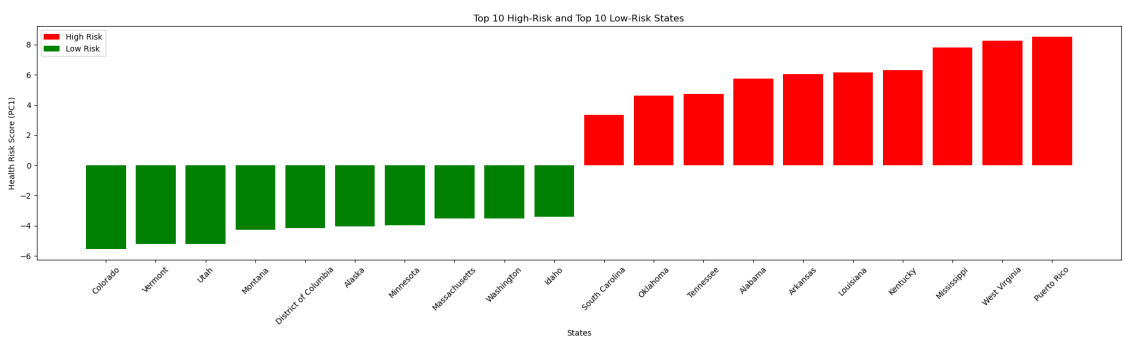
[85]:
```
plt.figure(figsize=(12,6))

cluster_df_sorted = cluster_df.sort_values("PC1")

plt.bar(cluster_df_sorted["LocationDesc"],
        cluster_df_sorted["PC1"],
        color=plt.cm.Set1(cluster_df_sorted["Cluster"]))

plt.xticks(rotation=90)
plt.title("State Health Risk Ranking")
plt.tight_layout()
plt.show()
```



0.0.4 Show Only Top & Bottom 10

[90]:
```
import matplotlib.pyplot as plt
from matplotlib.patches import Patch

legend_elements = [
    Patch(facecolor='red', label='High Risk'),
    Patch(facecolor='green', label='Low Risk')
]

# Top 10 high-risk
```

```
top10 = cluster_df.sort_values("PC1", ascending=False).head(10)

# Bottom 10 low-risk
bottom10 = cluster_df.sort_values("PC1").head(10)

combined = pd.concat([top10, bottom10])

# Sort combined for better visual order
combined = combined.sort_values("PC1")

# Map cluster numbers to colors
color_map = {0: "red", 1: "green"}
colors = combined["Cluster"].map(color_map)

plt.figure(figsize=(20,6))
plt.bar(combined["LocationDesc"],
        combined["PC1"],
        color=colors)

plt.xticks(rotation=45)
plt.xlabel("States")
plt.ylabel("Health Risk Score (PC1)")
plt.title("Top 10 High-Risk and Top 10 Low-Risk States")

plt.tight_layout()

plt.legend(handles=legend_elements)

plt.show()
```



<——————————— End of our First Exploratory Analysis ———————————
———->

—— **Regression Analysis Begins** ————

[106]:
```
df_cleaned_data["DataValueUnit"].value_counts()
```

[106]:
```
DataValueUnit
%                       203382
Number                   54104
cases per 100,000        31980
per 100,000              11680
cases per 1,000           7488
Years                      312
gallons                    165
cases per 1,000,000        104
Name: count, dtype: int64
```

[107]:
```
df_reg = df_cleaned_data.copy()

# Keep percent and rate-based indicators
df_reg = df_reg[
    df_reg["DataValueUnit"].isin(["%", "cases per 100,000", "per 100,000"])
].copy()
```

[108]:
```
wide_reg = df_reg.pivot_table(
    index=["LocationDesc", "YearStart"],
    columns="Question",
    values="DataValue"
).reset_index()

wide_reg.columns = wide_reg.columns.str.replace(" ", "_")
```

[109]:
```
state_reg = wide_reg.groupby("LocationDesc").mean(numeric_only=True).
 ↪reset_index()
```

[110]:
```
[col for col in state_reg.columns if "coronary" in col.lower()]
```

[110]:
```
['Coronary_heart_disease_mortality_among_all_people,_underlying_cause']
```

[111]:
```
target = "Coronary_heart_disease_mortality_among_all_people,_underlying_cause"
```

[112]:
```
print(target in state_reg.columns)
```

```
True
```

[113]:
```
predictors = [
    "Diabetes_among_adults",
    "High_blood_pressure_among_adults",
    "Current_smoking_among_adults",
    "No_leisure_time_physical_activity_among_adults",
    "High_cholesterol_among_adults_who_have_been_screened"
```

```
]
```

[114]:
```python
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np

X = state_reg[predictors]
y = state_reg[target]

# Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("R²:", r2_score(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[114], line 6
      3 from sklearn.metrics import r2_score, mean_squared_error
      4 import numpy as np
----> 6 X = state_reg[predictors]
      7 y = state_reg[target]
      9 # Split

File /opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages/pandas/
 ↪core/frame.py:4119, in DataFrame.__getitem__(self, key)
   4117     if is_iterator(key):
   4118         key = list(key)
-> 4119     indexer = self.columns._get_indexer_strict(key,          )[1]
   4121 # take() does not accept boolean indexers
   4122 if getattr(indexer, "dtype", None) == bool:

File /opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages/pandas/
 ↪core/indexes/base.py:6212, in Index._get_indexer_strict(self, key, axis_name)
   6209 else:
   6210     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 6212 self._raise_if_missing(keyarr, indexer, axis_name)
   6214 keyarr = self.take(indexer)
   6215 if isinstance(key, Index):
```

```
  6216     # GH 42790 - Preserve name from an Index

File /opt/conda/envs/anaconda-2025.12-py312/lib/python3.12/site-packages/pandas/
 ↪core/indexes/base.py:6264, in Index._raise_if_missing(self, key, indexer,␣
 ↪axis_name)
  6261     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
  6263 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
-> 6264 raise KeyError(f"{not_found} not in index")

KeyError: "['Current_smoking_among_adults',␣
 ↪'No_leisure_time_physical_activity_among_adults'] not in index"
```

[ ]:

```

```